Soyeon Kim

Ph.D. Candidate | Kim Jaechul Graduate School of Artificial Intelligence, KAIST

Seongnam, Korea | soyeon.k@kaist.ac.kr

GitHub | Google Scholar | ORCID

RESEARCH INTERESTS

Explainable and trustworthy AI for meteorology, geoscience, land use, and environmental systems

Interpretable forecasting models, concept-based explanations, and spatial decision support

Geospatial analysis and science-based evidence generation for environmental policy and planning

Weather AI agent for reliable, tool-augmented forecasting and decision-support workflows

Fine-tuning vision language models for meteorological interpretation and weather intelligence

Benchmarking and evaluating multimodal large language models for meteorological reasoning

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) | Seongnam, Republic of Korea

Ph.D. Candidate in Artificial Intelligence | Aug. 2020 - Present

Advisor: Jaesik Choi

Seoul National University (SNU) | Seoul, Republic of Korea

M.L.A. in Landscape Architecture | Mar. 2017 - Feb. 2019

Honors: Summa Cum Laude (GPA 4.24/4.30); Outstanding Master's Thesis Award

Thesis: Assessing Conservation Priority of Unexecuted Urban Parks Using the Urban Park Multifunctional Performance Index

Advisor: Yonghoon Son

Kangwon National University (KNU) | Gangwon, Republic of Korea

B.L.A. in Landscape Architecture | Mar. 2010 - Feb. 2016

Honors: Summa Cum Laude (GPA 4.35/4.50)

Advisor: Youngjo Yun

PROFESSIONAL EXPERIENCE

INEEJI Co. Ltd., AI Research Center | Seongnam, Republic of Korea

Principal Researcher | Dec. 2025 - Present

Led end-to-end meteorological AI projects spanning proposal development, forecasting support systems, weather-domain agents, a benchmark for evaluating multimodal LLMs in Korean meteorology, VLM fine-tuning, expert-informed evaluation design, and GPU resource acquisition through funded support programs.

Concurrent with Ph.D. studies since Dec. 2025.

National Institute of Ecology, Ecosystem Service Team | Seocheon, Republic of Korea

Research Fellow | Jun. 2019 - Aug. 2020

Conduct research on ecosystem service evaluation and mapping using GIS, R, and Python.

Design and analyze citizen and expert surveys to support evidence-based environmental policymaking.

Architecture and Urban Research Institute, Smart & Green Research Group | Sejong, Republic of Korea

Research Assistant | Oct. 2018 - Nov. 2018; Apr. 2019 - Jun. 2019

Researched smart-city and regulatory-sandbox case studies to support national land policy development.

PEER-REVIEWED PUBLICATIONS

1. **Soyeon Kim**, Kyowoon Lee, and Jaesik Choi. **"Diffusion Integrated Gradients: Controllable Path Generation for Flexible Feature Attribution."** *The 19th European Conference on Computer Vision (ECCV), 2026.*

Paper | Presentation | Poster

2. **Soyeon Kim**, Seongwoo Lim, Kyowoon Lee, and Jaesik Choi. **"Spectral Integrated Gradients for Coarse-to-Fine Feature Attribution."** *The 32nd ACM KDD Conference on Knowledge Discovery and Data Mining, Research Track, 2026.*

Paper | Poster | Code

3. **Soyeon Kim**, Seongwoo Lim, Kyowoon Lee, and Jaesik Choi. **"Manifold-Aligned Guided Integrated Gradients for Reliable Feature Attribution."** *The Forty-third International Conference on Machine Learning (ICML), 2026.*

Paper | Poster | Code

4. **Soyeon Kim**, Cheongwoong Kang, Myeongjin Lee, Eun-Chul Chang, Jaedeok Lee, and Jaesik Choi. **"K-MetBench: A Multi-Dimensional Benchmark for Fine-Grained Evaluation of Expert Reasoning, Locality, and Multimodality in Meteorology."** *Findings of the Association for Computational Linguistics, 2026.*

Paper | Leaderboard | Dataset | Code

5. Subeen Lee, Jiyeon Han, **Soyeon Kim**, and Jaesik Choi. **"Diverse Rare Sample Generation with Pretrained GANs."** *Proceedings of the AAAI Conference on Artificial Intelligence 39(5):4553-4561, 2025.*

Paper

6. **Soyeon Kim**, Junho Choi, Subeen Lee, and Jaesik Choi. **"Unsupervised Concept Discovery for Deep Weather Forecast Models with High-Resolution Radar Data."** *PLOS Climate 4(9):e0000633, 2025.*

Paper

7. **Soyeon Kim**, Junho Choi, Subeen Lee, and Jaesik Choi. **"Example-Based Concept Analysis Framework for Deep Weather Forecast Models."** *Artificial Intelligence for the Earth Systems 4(3), 2025.*

Paper

PEER-REVIEWED PUBLICATIONS - CONTINUED

8. **Soyeon Kim**, Jihyeon Seong, Hyunkyung Han, and Jaesik Choi. **"Capsule Neural Networks as Noise Stabilizer for Time Series Data."** *Journal of KIISE 51(8):678-684, 2024.*

Paper

9. **Soyeon Kim**, Junho Choi, Yeji Choi, Subeen Lee, Artyom Stitsyuk, Minkyoung Park, Seongyeop Jeong, Youhyun Baek, and Jaesik Choi. **"Explainable AI-Based Interface System for Weather Forecasting Model."** *HCI International 2023 - Late Breaking Papers, LNCS 14059:101-119, 2023.*

Paper

10. Boseon Yoo, Jiwoo Lee, Janghoon Ju, Seijun Chung, **Soyeon Kim**, and Jaesik Choi. **"Conditional Temporal Neural Processes with Covariance Loss."** *Proceedings of the 38th International Conference on Machine Learning (PMLR 139):12051-12061, 2021.*

Paper | Code

11. **Soyeon Kim**, Yonghoon Son, and Hajung Ko. **"Establishing Assessment Index of Prioritization of Building for Long-term Unexecuted Urban Parks: Considering the Urban Natural Park Area."** *The Study of Suwonology 14:77-105, 2019. In Korean. Research grant: KRW 2 million.*

Paper

12. Youngjo Yun and **Soyeon Kim**. **"Interspace Memorial Design in Urban Street: Fence Opening Design of The War Memorial of Korea's Plaza."** *Journal of Design & Arts 18:3-24, 2016. In Korean.*

MANUSCRIPTS UNDER REVIEW

1. **Soyeon Kim**, Seongwoo Lim, Kyowoon Lee, and Jaesik Choi. **"Robust Path Attribution in Piecewise Linear Neural Networks via Dummy-Constrained Optimization."** *Under review at the Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS), 2026.*

MANUSCRIPT IN PREPARATION

1. **Soyeon Kim**, Ilkwon Kim, and Hyeoksu Kwon. **"A Delphi-Matrix Approach to Environmental Impact Assessment Integrated with Ecosystem Services."** *Manuscript in preparation, 2020.*

SELECTED PRESENTATIONS

1. “What Can GIS Do to Save Urban Parks That Will Disappear All at Once? Prioritizing Installation and Cancellation of Long-Term Unexecuted Urban Parks (LUUP).” Esri User Conference, San Diego, CA, USA, 2018. Young Scholar Award.

2. “Explainable Artificial Intelligence for Precipitation Forecasting.” 60th Conference of the Korea Meteorological Society, Busan, Republic of Korea, 2023. Oral presentation.

3. “Explainability and Variable Importance of Deep Learning-Based Short-Term Precipitation Forecasting Model.” 60th Conference of the Korea Meteorological Society, Busan, Republic of Korea, 2023. Oral presentation.

4. “Which Explanation Is the Best? A Systematic Evaluation of Feature Attribution Methods on Precipitation Forecasting Model.” 60th Conference of the Korea Meteorological Society, Busan, Republic of Korea, 2023. Poster.

5. “User-Centric XAI Interface System for Weather Forecasting AI.” 60th Conference of the Korea Meteorological Society, Busan, Republic of Korea, 2023. Poster.

6. “Capsule Neural Networks as Noise Stabilizer for Time Series Data.” Korea Computer Congress, Jeju, Republic of Korea, 2023. Oral presentation. Best Paper Award.

7. “The Land Usage Types of Suburban Forest Boundaries.” Autumn Conference of the Korean Society of Environmental Landscape Architecture, Seoul, Republic of Korea, 2017. Oral presentation. Outstanding Presentation Award.

8. “Zoning Classes for Forest Visual Landscape Management in and around the Pyeongchang Winter Olympic Area: Focused on the Alpensia District, Pyeongchang, Gangwon-do.” Conference of the Korea Society of Environmental Restoration Technology, Seoul, Republic of Korea, 2013. Oral presentation.

HONORS AND AWARDS

ACM SIGKDD Artifact Badge, for research artifact availability and reproducibility associated with “Spectral Integrated Gradients for Coarse-to-Fine Feature Attribution,” KDD 2026 Artifact Badging, 2026

CVPR Compute Transparency Champion Award, for compute reporting associated with “Integrated Gradients with Diffusion for Flexible Input Attribution,” CVPR 2026 Compute Reporting Initiative, 2026

Graduation with Honors (Summa Cum Laude) and Outstanding Master’s Thesis Award, Seoul National University, 2019

Young Scholar Award, Esri User Conference, 2018

Japan Student Services Organization (JASSO) Scholarship for Short-Term Study, Chiba University, 2018

Outstanding Presentation Award, Autumn Conference of the Korean Environmental Landscape Architecture Society, 2017

Honorable Mention, Academic Paper Contest on the Quality of Life of Gyeonggi-do Residents, 2017

Graduation with Honors (Summa Cum Laude), Kangwon National University, 2016

National Science and Engineering Scholarship (Full Tuition), Korea Student Aid Foundation, 2014-2015

Academic Scholarship (Full Tuition), Kangwon National University, 2012-2013

PROFESSIONAL SERVICE

Journal Reviewer: IEEE Transactions on Neural Networks and Learning Systems (2020)

Conference / ARR Reviewer or Subreviewer: NeurIPS (2023, 2025), ICLR (2023), UAI (2024), ACML (2024), ACL ARR (2025), SIGKDD (2026)